SovereignNexus Master Receipt & Provenance Log

Axiom: 1=1=1 | **Status:** Active Audit Ledger | **Classification:** Verified Architectural Baseline

1. External Benchmarks & Literature Citations

The following peer-reviewed literature citations establish physical hardware parameters and thermodynamic boundaries for the SovereignNexus substrate architecture.

[REF-01] Quantum Substrate Baseline

Target Section: Section §7.8 (Diamond Substrate & Coherence)

Source: Yamamoto et al. (2026)

Publication: *npj Quantum Information*

DOI: 10.1038/s41534-026-01368-y

Application: Anchors the 13 ppm ^{13}C isotopic purification threshold and 11.2 s CPMG room-temperature spin coherence target specification.

[REF-02] Thermal & Material Governance

Target Section: Section §7.5 (Thermal Extraction & Landauer Governor)

Source: CVD Diamond Thermal Properties & Boundary Resistance Literature

Publication: Peer-Reviewed Heat Transfer & Materials Physics

Application: Validates thermal conductivity (>2000 W/mK) and GaN-on-Diamond heterostructure heat dissipation limits.

** Log updated continuously as architectural modules are verified and anchored.*